Appendix A: Maintenance & Troubleshooting

This appendix covers the following topics and tasks:

- ✓ Troubleshooting
- ✓ Maintaining and servicing your gauge
- ✓ Charging the battery
- ✓ Replacement parts list
- ✓ Returning the gauge for service

Troubleshooting

Gauge Fails Standard Counts

Ensure that the source rod opening on the gauge bottom is completely closed or covered by the tungsten sliding block. If any opening is visible, the sliding block should be cleaned by the Radiation Safety Officer as described in Appendix D. If the sliding block still does not close completely, contact the nearest Troxler Service Center.

Ensure that the guidelines for performing the standard count listed on page 3-9 are followed.

Perform the standard count again. If it still fails, contact your nearest Troxler service center or representative.

No Density Readings

The most likely reason for no density readings is an electronic problem, such as a failure of the detector preamplifier.

Also, the gauge may have gotten wet. The Model 3430 and 3440 gauges are not waterproof and should not be used in the rain. If the gauge gets wet make sure it is completely dry before sealing it in the case for storage. If you suspect that moisture may be inside the gauge open the keypad panel and run a fan or hairdryer (on low) in the gauge for 15 minutes to an hour in a clean environment. .

As a precaution, ensure that the tip of the source rod is intact and undamaged. Use a radiation survey meter to check the radiation levels at the surface of the gauge base where the source rod exits (without extending the source rod). A maximum reading of 10-20 mrem/hr is normal, and indicates the source is present. If the maximum reading is less than 1 mrem/hr or if a survey meter is not available, **perform a visual inspectio**n of the source rod tip as follows to confirm its integrity.

Extend the source rod just far enough to see the source rod tip. The tip should appear flat to slightly rounded and smooth.

Complete the inspection as quickly as possible and stay at least 1 meter from the rod tip to minimize exposure (the dose rate at 1 meter from the unshielded source is about 2.7 mrem/hr).

If the visual inspection indicates that the source rod tip is broken off (source is missing):

1. Immediately contact your Radiation Safety Officer (RSO).
2. Initiate a search for the source starting at the location where the gauge was last used.
3. Report lost or missing radioactive sources to your state or federal radiation control agency in accordance with applicable regulatory requirements.
4. Contact the Troxler Radiation Safety Department for further advice.

Gauge Readings Appear Erratic

1. Ensure that the source rod is properly positioned in the desired measurement position. For backscatter readings, do not allow the source rod to contact the material.
2. Erratic density readings may be caused by a dirty sliding block. Clean the sliding block as instructed in Appendix D and then take readings again.
3. Check the inside of the gauge for moisture. To dry the gauge interior, remove the keypad. If necessary, use a hair dryer (on low heat) to circulate warm air for 15 minutes to 1 hour.
4. Remove any foreign objects from inside the gauge.
5. Ensure the hardware mounting screws are tight and in place.
6. Check count time – a four-minute count will give the highest precision with a repeatability of ±1 pcf.
7. Perform a statistical stability (stat)test.
   - ▶ If test passes, proceed with job.
   - ▶ If test fails, repeat two more times. If test fails two out of three times, contact the nearest Troxler Service Center.

NOTE

To aid in verifying gauge readings, after a gauge has been calibrated, mark a test area on a concrete floor, sidewalk, or equivalent and measure the density (WD). This measurement can then be used as a reference to verify later gauge readings.

Unreadable, XXXX or ++++ is Displayed

1. Check the standard counts in memory. If the standard counts are suspect, perform new standard counts. If counts equal zero for both systems, replace high voltage board. Contact the nearest Troxler Service.
2. Ensure that the measurement depth (depth of the source rod handle) is the same as the depth displayed on the display before starting a measurement.
3. Check gauge for water damage. If the gauge is wet, dry the gauge interior with hairdryer (on low heat) for 15 minutes to 1 hour..
4. Check that the calibration constants are valid.
5. If necessary, perform a statistical stability (stat) test, record the results and contact the nearest Troxler Service Center.

Gauge Turns off after it is Turned On

1. The gauge automatically turns off after five hours if no keys are pressed. Try to turn the gauge on again.
2. The gauge may be wet. Do not turn the gauge on until moisture is removed from gauge interior! Component damage may result.
3. If the battery is below 5.5 volts, recharge or replace the batteries.
4. The front panel assembly may be defective. To test, replace the suspect front panel assembly with a good front panel assembly.

Short Battery Life after Recharging

1. Check that you are using the correct charger.
2. Ensure the charging indicator light (red LED) is illuminated when the gauge is plugged in.
3. NiMH batteries may be charged up to 500 full charge-discharge cycles. The batteries may be reaching end of life cycle - replace. Note that all information stored in the gauge except the calibration constants and the chosen language is lost when the batteries are disconnected.
4. Remove any loose screws or foreign objects from the gauge interior that may cause an electrical short to ground.
5. The AC charger may be defective. Check voltage output of charge with a voltmeter or use the dc charger to charge the batteries.

Satisfactory Counts, but Results are in Error

- Ensure that the handle is seated in bottom of notch.
- Ensure the measurement depth on the display screen corresponds to the actual source rod handle depth.
- Check to see if an offset (density, moisture, trench or special) is enabled.
- Ensure that the standard counts are correct.
- Check calibration constants.

Possible Malfunction Indicators

Symptom	Possible Malfunction
Batteries discharge prematurely	HV Board
Batteries do not charge	CPU Board, Preamp Board
Battery low indicator does not function correctly	CPU Board
Beeper stops (or is erratic)	CPU Board
Display malfunctions	CPU Board
Display test Fails	CPU Board
Fails stability or drift tests	Preamp board
Gauge does not turn on when charger is connecte	CPU Board
Gauge doesn't turn off	CPU Board
Gauge fails stat or drift tests	HV Board
Gauge fails tube test	Preamp board
Moisture or density counts are unstable	HV Board
No keypad response	CPU Board
No moisture or density counts	Preamp Board, HV Board

Error Messages

The following error messages are not user-serviceable. Contact Troxler Customer Service for more information.

- KEY PAD TEST ERROR!
- GM TUBE TEST ERROR!
- HELIUM TUBE TEST ERROR!
- DISPLAY TEST ERROR!

Battery Charging

With fully charged batteries, the Model 3430 gauge will remain operational for approximately eight weeks under normal (8-hour day) conditions.

If the batteries become discharged, the following message will be displayed on the gauge:



When this display appears, there are a few hours remaining before the battery must be recharged. In an emergency, a 30-minute recharge with the dc or ac charger gives several hours of use. If possible, run the batteries down before recharging.

NOTE

Batteries should not be recharged unless the Battery Low! **indication is displayed!**

Installing and Using Alkaline Batteries

If your Model 3430 gauge is installed with the optional alkaline battery backup (see page 5–3), you can power the gauge using five AA alkaline batteries. To install the batteries:

1. Turn the gauge off.
2. Loosen the four captive screws on the control unit.
3. Carefully lift the control unit from the gauge.
4. Unplug the ribbon cable from the control unit by pushing the levers on either side of the ribbon down.
5. Install five standard AA alkaline batteries into the battery holder.
6. Set the battery select switch to the **ALKALINE** position.
7. Plug the ribbon cable into the control unit.
8. Attach the control unit to the gauge with the fourcaptive screws.

Mechanical Maintenance

The following procedures should be performed to keep the Model 3430 gauge in good working order. If a serious problem with the gauge arises, contact the nearest Troxler Service Center or representative for instructions. For a list of Troxler and authorized Troxler service centers, refer to the front of this manual or visit the Troxler website at www.troxlerlabs.com

⚠ CAUTION

Personnel should wear a dosimeter to monitor radiation exposure while performing maintenance on the gauge if it is required by their radioactive materials license or local regulations.

Cleaning

If the Model 3430 gauge is to provide precise and accurate measurements over a long period of time the gauge should be kept as clean as possible. Monitor the outside surfaces of the instrument for accumulations of dirt, oil, asphalt, or any other foreign matter. If a build-up of material is visible on the gauge base or topshell, use the following procedures for cleaning:

To clean the gauge base, use a putty knife to scrape away any built-up accumulations of soil or asphalt. **Be careful not to damage the gauge base!** After removing any large accumulations, wipe the gauge base with a cloth soaked in BindOff, WD-40 or similar non-corrosive cleaner to remove the remaining debris.

CAUTION

Cleaners and lubricants can damage the keypad. Do not allow these chemicals to make contact with the keypad!

Currently, the Model 3430 gauge topshell is manufactured from an engineering thermoplastic designed specifically to provide

high impact strength and to offer excellent compatibility with many industrial solvents and petrochemicals. The topshell may be cleaned with mild (low alkaline) soap and water. Other approved cleaning substances include methyl, isopropyl, or isobutyl alcohols.

⚠ CAUTION

The use of any unapproved cleaning agents such as methyl-ethyl-ketones, amines, and methylene chloride will damage the topshell and void the warranty.

Source Rod Lubrication

If the source rod does not slide up and down freely, the source rod bearing may require cleaning and lubrication.

- Remove the control panel assembly from the gauge by loosening the four captive screws that secure it to the topshell. Lift the control unit and disconnect the ribbon cable. Set control unit aside.
- Locate the grease fitting at the base of the source rod tower.

NOTE

The vent valve was not installed on earlier gauges. Instead, an Allen-head screw was located beneath the grease fitting. On the gauges, remove the Allen-head screw before applying lubricant to allow the old grease to be ejected. Failure to remove the Allen screw may result in severe mechanical damage to the base assembly.

- Using a standard 16-ounce grease gun loaded with a Magnalube-G cartridge, apply five shots of lubricant or enough to eject all dirty grease until clean grease is visible.
- Re-assemble the gauge. Do not apply more than 6 in-lb torque to the topshell screws.

Gasket Replacement

Four gaskets seal the gauge from moisture, dirt, and debris. Condensation may form inside the gauge if the gauge has been out in extremely wet weather or in high humidity, or if it is used in the heat and stored in an air conditioned building. In this case, remove the topshell and allow the gauge to dry. If necessary, use a hairdryer (on low heat) to remove any built-up moisture. If moisture continues to be a problem or debris is present inside the gauge base, the gaskets may need replacing.

To replace the control panel gasket (P/N 110841), loosen the four screws in the corners of the panel, and remove the panel from the gauge topshell. Use care in removing the panel and disconnecting the ribbon cable. Gently peel the old gasket from the panel and replace with a new gasket. Reconnect the ribbon cable and replace the control panel.

To replace the two post gaskets and the topshell gasket, remove the screws that hold the topshell to the gauge base. Remove the screws from the metal triangle around the base of the triangle extrusion and raise this piece. Then lift the topshell from the base. Gently peel the gaskets away from the topshell and gauge base and replace. Tighten topshell screws to 6 in-lb.

If replacing the post gaskets, slide up and over the source rod handle. A light coating of talcum powder on the inside of the new post gasket will aid in reassembly.

Replacement Parts

This section provides a list for replacing the major parts of the gauge and purchasing accessories. Many parts can be ordered from our e-commerce site at www.troxlerlabs.com.

PART NO.	DESCRIPTION
110899	3430 Gauge Topshell O-ring/Gasket
110013	3430 Bottomplate
110016	3430 Battery Assembly (with case)
110821	3430 Topshell
110848.0040	3430 Nameplate
007158	O-ring for Triangle Tower

Maintenance Supplies

PART NO.	DESCRIPTION
012784	Lubricant, Magnalube-G paste 1.5 oz tube
012786	Lubricant, Magnalube-G paste 1 lb can
012789	Lubricant, Magnalube-G 14.5 oz
100761	Source rod pig
018141	BindOff (1 gallon)

Accessories

PART NO.	DESCRIPTION
110017	3430P/3440 Shipping Case
100421	Drill Rod
102111	Scraper Plate
103680.1000	Extraction Tool
110403	AC Adapter
104156	DC Charger
110990	3430 Operator's Manual
113128	RS-232 Cable
102876.0005	Leak Test Kit (4 units)
109661	TroxAlert Survey Meter
104661.4000	Printer Package

Returning the Gauge for Service

All shipments within the United States to the factory must be accompanied by an RGA (Returned Goods Authorization) number, and a description of the instrument and its problem. Send a completed RGA form, found on the Downloads page at www.troxlerlabs.com, with each item returned for service. This information is used by Troxler shipping and service personnel to expedite the repair work.

To obtain an RGA number, please call or fax the factory or branch office with your request. Please have the following information available when contacting Troxler for an RGA number:

- Is the gauge still under warranty?
- Model and Serial number
- Will estimate be required before performing any work on the gauge?
- Problem or difficulty you are having with the instrument.
- Shipment method to Troxler and for return shipment.
- Shipping and billing address (not P.O. Box): street address and zip code.
- Telephone number and contact (for questions from Troxler).
- Payment method: credit card, account number, or purchase order number. All U.S. government agencies (city, county, state and federal) must send purchase orders.

Appendix B: Specifications

This appendix provides the measurement specifications for Model 3430 gauges.

Measurement Specifications

Density at 2000 kg/m3

Direct Transmission (150 mm)

	.25 min	1 min	4 min
Precision (kg/m^3)	±6.8	±3.40	±1.70
Composition error (kg/m^3)	±20.0	±20.0	±20.0
Surface error (kg/m^3) (100% Void)	–17.0	–17.0	–17.0

Backscatter (98%, 100 mm)

	.25 min	1 min	4 min
Precision (kg/m^3)	±16.0	±8.00	±4.00
Composition error (kg/m^3)	±40.0	±40.0	±40.0
Surface error (kg/m^3) (100% Void)	–75.0	–75.0	–75.0

Moisture at 240 kg/m^3

	.25 min	1 min	4 min
Precision (kg/m^3)	±10.3	±5.1	±2.6
Surface error (kg/m^3) (1.25 mm, 100% void, kg/m^3)	–18.0	–18.0	–18.0

Precision is defined as ±one (1) standard deviation in density readings. This number is calculated by the ratio of the standard deviation in the counting rate and the slope of the calibration curve at a given density.

U.S. Customary Units

Density at 125 PCF

Direct Transmission (6 inches)

	.25 min	1 min	4 min
Precision (pcf)	±0.42	±0.21	±0.11
Composition error (pcf)	±1.25	±1.25	±1.25
Surface error (pcf) (100% Void)	–1.06	–1.06	–1.06

Backscatter (98%, 4 inches)

	.25 min	1 min	4 min
Precision (pcf)	±1.00	±0.50	±0.25
Composition error (pcf)	±2.50	±2.50	±2.50
Surface error (pcf) (100% Void)	–4.68	–4.68	–4.68

Moisture at 15 PCF

	.25 min	1 min	4 min
Precision (pcf)	±0.64	±0.32	±0.16
Surface error (pcf) (0.05 in., 100% void, pcf)	–1.12	–1.12	–1.12

Radiological Specifications

Gamma Source	0.30 GBq (8 mCi) ± 10% Cs-137
Neutron Source	1.48 GBq (40 mCi) ± 10% Am-241:Be
Source Type	Sealed Source – Special Form
Source Housing	Stainless Steel, Encapsulated
Shielding	Lead, Tungsten
Surface Dose Rate	See Radiation Profile on page D-18
Shipping Case	Type A, Yellow II, TI = 0.3

Electrical Specifications

Stored Power	4 ampere hours

Average Current Consumption

Normal	35 mA
With backlight on	140 mA
With GPS on	105 mA
With backlight and GPS on	210 mA

Time Before Automatic Shutdown	5 hours of complete inactivity
Power Source(s):	
Main	5 C NiMH (Rechargeable Pack) batteries

Backup (optional)	5 AA alkaline batteries
Charge Source	12 V dc, 2A
Battery Recharge Time	3 hours maximum, automatic cutoff (may be charged incrementally without damaging the batteries)
Liquid Crystal Display	Alphanumeric, 4 line × 20 character, backlit
Keypad	10-key sealed membrane
Serial Data Format	9600 baud 1 stop bit No parity 8 data bits Xon-Xoff flow control

Gauge-to-PC Computer Cable
(Null Modem Serial Cable, Part Number 113128)

9 pin FEMALE	9 pin FEMALE
Rx (pin 2)	Tx (pin 3)
Tx (pin 3)	Rx (pin 2)
Gnd (pin 5)	Gnd (pin 5)

USB Port (optional)	A list of compatible USB devices is available at the Documents page of Troxler's website (www.troxlerlabs.com)

Mechanical Specifications

Gauge Size (w/ handle)	
12-Inch Rod	597 H × 368 L × 229 W mm (23.5 H × 14.5 L × 9 W in.)
8-Inch Rod	495 H × 368 L × 229 W mm (19.5 H × 14.5 L × 9 W in.)
Shipping Case Size	745 H × 419 L × 353 W mm (29.35 H × 16.5 L × 13.88 W in.)
Weight	13.8 kg (30.5 lb.)
Shipping Weight	37.6 kg (83.0 lb.)
Operating Temperature	0 to 70 °C 32 to 158 °F
Max Test Material Surface	175 °C (347 °F) for 15 minutes
Storage Temperature	–55 to 85 °C –67 to 185 °F
Humidity	98% RH, non-condensing

⚠ CAUTION

This instrument contains sensitive electronic and nuclear components. This instrument *must not* be subjected to stress, abuse, or use other than in accordance with the standard operating procedures listed in this manual.

Appendix C: Transporting & Shipping

Devices containing radioactive materials must be transported in accordance with the rules of the U.S. Department of Transportation (DOT) and the International Atomic Energy Agency (IAEA).

The IAEA recommendations have been codified in the International Air Transport Association (IATA) Dangerous Goods Regulations. International customers should consult their local government or licensing authority for applicable regulations.

U.S. Shipping Requirements

The U.S. DOT hazmat regulations (49 CFR, Parts 100–185) apply any time a nuclear device is transported by motor vehicle on a public highway or by other means of transport (rail, air, ship).

The major requirements for transporting a nuclear gauge in the United States are listed below. For more detailed information about these requirements, please refer to the *Troxler Transportation Guide*.

- A copy of the current IAEA Certificate of Competent Authority for each source in the gauge (Special Form Certificate) must be kept on file. Current versions can be downloaded from the Troxler website, www.troxlerlabs.com.
- A copy of the results of the Type A package testing must be kept on file.
- Hazmat employee training records must be kept on file.
- An *Emergency Response Information* document must be in the vehicle and immediately accessible to the driver.
- A properly completed bill of lading must be in the vehicle and immediately accessible to the driver. The shipping papers must include a 24-hr emergency response phone number.
- If shipping by air, a *Shipper's Declaration for Dangerous Goods* must accompany the air waybill.
- The package must be properly marked and labeled in accordance with hazmat regulations.
- The package must have a tamper-evident seal.
- The package must be inspected prior to each shipment.
- The package must be securely blocked and braced in the vehicle to prevent shifting during transport.

Accident Notification Requirements

In the event of a reportable incident involving radioactive material, notify the licensing agency as soon as practical. The operator is also required to notify, at the earliest practical moment, the U.S. DOT at 1-800-424-8802 of an accident that occurs during the course of transportation (including loading, unloading, and temporary storage) in which fire, breakage, spillage, or suspected contamination occurs involving shipment of radioactive materials.

Hazmat Training

The U.S. DOT regulations require every hazmat employer to train, test, certify, and maintain records for each hazmat employee. Hazmat training applies to anyone who transports or prepares for transport radioactive materials. Refresher training is required every three years for shipment / transport by roadway in the US. Shipment by air requires training every 2 years under IATA Dangerous Goods regulations.

Canadian Shipping Requirements

The *Transportation of Dangerous Goods Act and Regulations* (TDG) and *Transport Packaging of Radioactive Materials Regulations* (TPRM) apply any time a nuclear device used in commerce is transported by any means in Canada.

For training and accident notification requirements, consult the *Transportation of Dangerous Goods Regulations*. For further information on transporting a nuclear device, contact the transportation section of the Canadian Nuclear Safety Commission (CNSC).

Appendix D: Radiation Theory & Safety

This appendix is required reading for anyone who will operate the Model 3430 Surface Moisture- Density Gauges.

This appendix covers topics related to radiation theory and the safe operation of the gauge. A brief overview of the regulatory requirements related to the ownership and use of the gauge, as well as a listing of radiation safety-related warnings and cautions, is included.